

2019 HKDSE Physics

Paper 1B — Structured Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Editable transcription of all question text, formulae, symbols and units. Only essential original figures are retained; repeated figure content is omitted.

Official answers are placed directly below each question.

Question 1

- (a) An insulated container of negligible heat capacity contains 1.5 kg of tea at 60 °C.
- (i) What mass of ice at 0 °C should be added so that the final temperature is 10 °C? Assume the specific heat capacity of tea equals that of water. (3 marks)
- Given: specific latent heat of fusion of ice = $3.34 \times 10^5 \text{ J kg}^{-1}$; specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$.
- (ii) If the heat capacity of the container were not negligible, would more, less or the same mass of ice be required? Explain. (2 marks)
- (b) Ice cream at $-10 \text{ }^\circ\text{C}$ is placed in a thermal bag whose inner layer is polyethylene foam coated with aluminium foil. The bag has a zipper at the top and is taken outdoors on a hot sunny day.
- (i) With reference to heat-transfer processes, explain ONE feature that helps keep the ice cream at a low temperature. (1 mark)
- (ii) Suggest ONE modification that would improve the bag's ability to keep its contents cold. (1 mark)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks												
1. (a) (i) $(1.5)(4200)(60 - 10) = m(3.34 \times 10^5) + m(4200)(10 - 0)$ $m = 0.837766 \text{ kg} \approx 0.838 \text{ kg}$ (Accept: $m = 0.83 \sim 0.84 \text{ kg}$)	1M+1M 1A	All 3 terms correct 2 M Any 2 terms correct 1 M ΔT must be correct												
	3													
(ii) More ice is needed to cool the container as it will release heat/thermal energy as well.	1A 1A	<table border="1"> <thead> <tr> <th>Reason</th><th>More ice</th><th>Less ice</th></tr> </thead> <tbody> <tr> <td>✓</td><td>2</td><td>0</td></tr> <tr> <td>×</td><td>0</td><td>0</td></tr> <tr> <td>incomplete</td><td>1</td><td>0</td></tr> </tbody> </table>	Reason	More ice	Less ice	✓	2	0	×	0	0	incomplete	1	0
Reason	More ice	Less ice												
✓	2	0												
×	0	0												
incomplete	1	0												
	2													
(b) (i) Conduction: - Foam is a poor conductor (of heat) and it minimizes the transfer of heat/thermal energy from the surroundings to the cool contents/ice cream (inside the bag). OR Convection: - The zipper prevents convection between the hot air outside and the cool contents/ice cream (inside the bag).	Any ONE 1A	Key word: heat / energy One feature + corresponding reason OR Radiation: - The shiny surface inside the bag reduces the transfer of heat/thermal energy from the surroundings to the cool contents/ice cream (inside the bag) through emission of radiation												
	1													

(a)(i) Energy lost by tea = energy to melt and warm ice: $(1.5)(4200)(60-10) = m(3.34 \times 10^5) + m(4200)(10-0)$, giving $m = 0.838 \text{ kg}$.

(ii) More ice is required if the container has non-negligible heat capacity because the container also releases thermal energy while cooling to 10°C .

(b)(i) The foam is a poor conductor, the zipper suppresses convection, or the shiny foil reduces radiative transfer.

(ii) A valid improvement is a shiny outer surface or a thicker insulating layer.

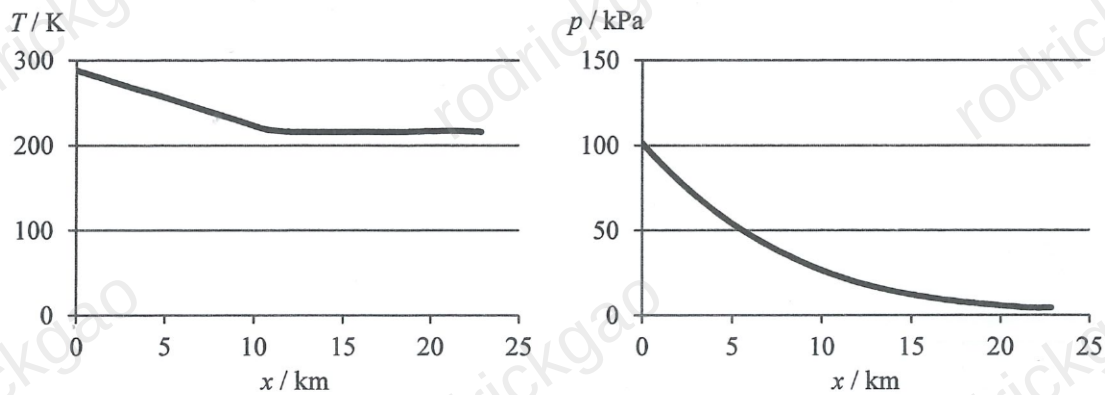
Question 2

A weather balloon of volume 0.52 m^3 contains helium at 15°C and 100 kPa at ground level.

(a) Find the amount of helium gas, in mol. (2 marks)

(b) The graphs show how air temperature T and atmospheric pressure p vary with height x . Assume the helium has the same temperature and pressure as the surrounding air at each height.





- (i) A student believes that because air temperature decreases over the first 10 km, the balloon volume decreases. Using the graphs, explain qualitatively why this is incorrect. (2 marks)
- (ii) From 12 km upward the temperature is steady at 216 K. When the balloon volume reaches 8 m^3 : (1) estimate the gas pressure; (2 marks) (2) hence find the height, using $p = p_0 e^{-kx}$, where p_0 is ground-level pressure and $k = 0.138 \text{ km}^{-1}$. (2 marks)

Official marking scheme

		(inside the bag).
(ii)		(Radiation) Make the outer surface (of the bag) shiny.
2. (a)		$\frac{pV}{T} = nR$ $(100 \times 10^3)(0.52) = n(8.31)(273+15)$ $n = 21.727504 \text{ (mol)} \approx 21.7 \text{ (mol)}$
(b) (i)		Since $pV = nRT \Rightarrow V = \frac{nRT}{p}$ / volume V of the balloon depends on both T and p , the (fractional) decrease in pressure p (with height) is greater/faster than the (fractional) decrease in temperature T .
(ii) (1)		$\frac{pV}{T} = \text{constant}$ $\frac{(100)(0.52)}{(273+15)} = \frac{p(8)}{216}$ $p = 4.875 \text{ kPa or } 4875 \text{ Pa}$
(2)		$p = p_0 e^{-kx}$ $4.875 = 100 e^{-0.138x}$ $x = 21.89166726 \text{ (km)} \approx 21.9 \text{ (km)}$
		bag) through emission of radiation
	1A	The answer must be related to the modification of the bag like 'thickening the bag'
	1	Adding ice to the bag is NOT accepted
	1M	Accept: $n = 21 \sim 22 \text{ mol}$
	1A	Accept: volume V decreases as temperature T decreases would be true only if pressure p is constant, however, p decreases with height.
	2	
	1M	Note: $p(8) = 21.7 \times 8.31 \times 216$
	1A	$p = 4869 \text{ Pa}$
	2	
	1M	e.c.f. from (b)(ii)(1)
	1A	Accept 21.8 km to 22.0 km
	2	

(a) $pV = nRT$ gives $n = (100 \times 10^3)(0.52) / [8.31(288)] = 21.7 \text{ mol}$.

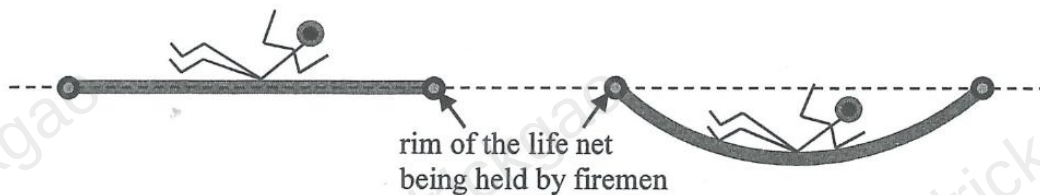
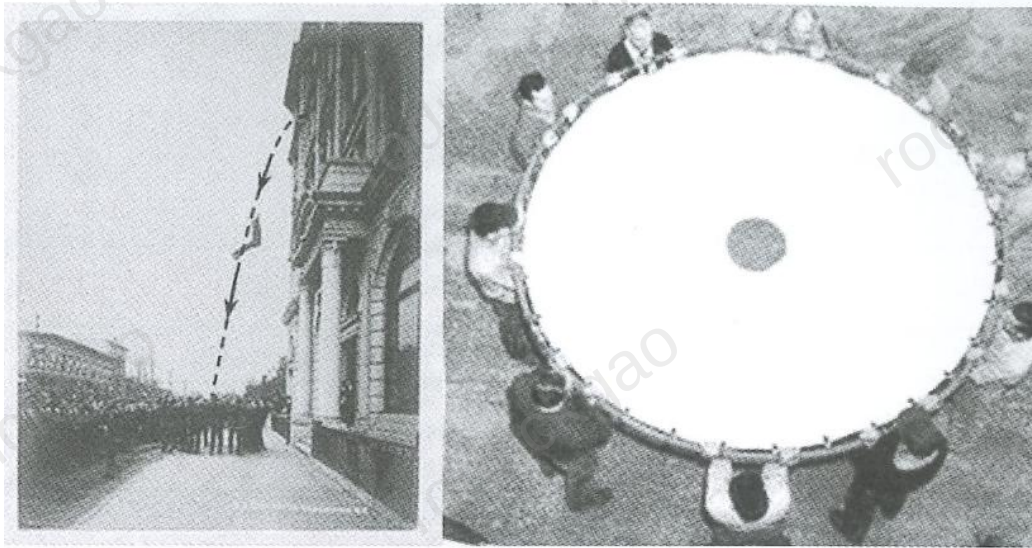
(b)(i) $V \propto T/p$; the fractional fall in pressure is greater than that in temperature, so the balloon expands.

(ii) (1) From $p_1 V_1 / T_1 = p_2 V_2 / T_2$, $p_2 = 4.875 \text{ kPa}$. (2) With $p = 100 e^{-0.138x} \text{ kPa}$, $x = 21.9 \text{ km}$.

Question 3

Read the following passage about life nets and answer the questions.

A life net was formerly used by firefighters to allow people on upper floors of a burning building to jump to safety. Its practical height limit was about six storeys, although some people survived jumps from an eight-storey building with varying injuries. When a person hits the net, it deforms and brings the person to rest over a longer time than a solid ground would.



- (a) A person falls from 12 m above a life net with negligible initial speed. Neglect air resistance and the person's size. ($g = 9.81 \text{ m s}^{-2}$)
- Estimate (1) the vertical speed v and (2) the time of fall t just before the person reaches the net. (4 marks)
 - If the 70 kg person is stopped in 0.3 s, estimate the average force exerted on the person by the net. (3 marks)
 - What form of energy is stored by the life net while the falling person is being decelerated? (1 mark)
- (b) (i) Give a reason why there is a height limit for using life nets. (1 mark)
- (ii) A falling person might hit the rim, injuring the person or the firefighters. Explain why it is not easy for a person jumping from a height to reach the central part of the net. (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
3. (a) (i) (1) $\frac{1}{2}mv^2 = mgh$ $v^2 = 2(9.81)(12)$ $v = 15.344054 \text{ m s}^{-1} \approx 15.3 \text{ m s}^{-1}$ ($v = 15.491933 \text{ m s}^{-1} \approx 15.5 \text{ m s}^{-1}$ for $g = 10 \text{ m s}^{-2}$)	1M 1A 2	Accept: $v = 15 \sim 15.5 \text{ m s}^{-1}$
(2) $s = \frac{1}{2}gt^2$ $12 = \frac{1}{2}(9.81)t^2$ $t = 1.564124 \text{ s} \approx 1.56 \text{ s}$ ($t = 1.5491933 \text{ s} \approx 1.55 \text{ s}$ for $g = 10 \text{ m s}^{-2}$)	1M 1A 2	
(ii) $F - mg = ma$ $F = \frac{70 \times (15.3 - 0)}{0.3} + 70 \times 9.81$ $= 4266.9793 \text{ N} \approx 4270 \text{ N}$ ($F = 4314.7845 \text{ N} \approx 4310 \text{ N}$ for $g = 10 \text{ m s}^{-2}$)	1M+1M 1A 3	$F = ma$ 1 M $F = ma + mg$ 2 M Accept: $F = 4180 \sim 4320 \text{ N}$
(iii) Elastic potential energy	1A 1	MUST have the word: 'Elastic' Accept: Elastic energy
(b) (i) (Velocity is too high, hence the force for deceleration is too large.) - The life net may be torn. - The falling person may be injured. - The firemen may not be able to hold the life net tight.	1A Any ONE	The answer is related to the consequence due to limitation of any one of the following: (1) the net, (2) the falling person (3) the firemen NOT accept: - The force is too large which exceed the limit of the life net - The life net is not strong enough to withstand the force - The force is too large that the man cannot withstand
(ii) There exists a <u>horizontal velocity</u> when a person jumps and the horizontal displacement is very difficult to estimate as it depends on the time of fall, which is usually long.	1A 1A 2	1 st mark: horizontal velocity / projectile motion 2 nd mark: difficult to estimate the horizontal displacement / horizontal direction

(a)(i) $v = \sqrt{2gh} = 15.3 \text{ m s}^{-1}$ and $t = \sqrt{2h/g} = 1.56 \text{ s}$.(ii) Upward net force gives $F - mg = m\Delta v/\Delta t$, so $F = 70(15.3/0.3) + 70g \approx 4.27 \text{ kN}$.

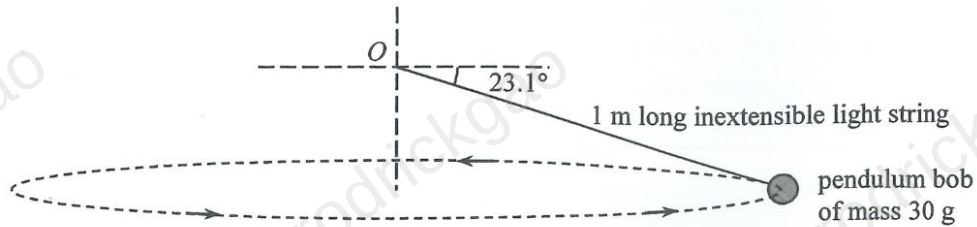
(iii) The net stores elastic potential energy.

(b)(i) From excessive height, impact speed and stopping force may injure the person/firefighters or tear the net.

(ii) A jumper has uncertain horizontal velocity; with a long fall time, the resulting horizontal displacement is difficult to predict.

Question 4

- (a) A 30 g pendulum bob is tied to O by a 1 m light inextensible string that makes an angle 23.1° below the horizontal. It describes a horizontal circle uniformly at angular velocity 5.0 rad s^{-1} . Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)



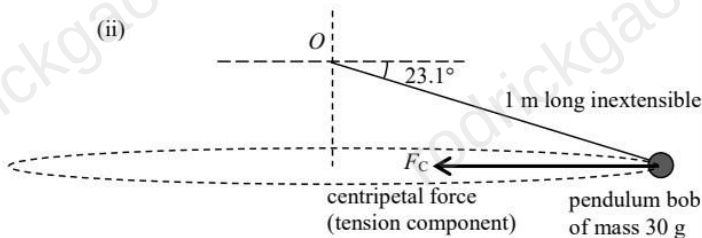
- (i) What is the rotation rate in revolutions per second? (1 mark)
- (ii) Indicate the centripetal force F_c required and find F_c . (3 marks)
- (iii) Explain whether the string tension is greater than, equal to or smaller than F_c . (2 marks)
- (b) The Moon moves uniformly in a circular orbit around Earth under Earth's gravitational attraction.
- (i) Explain why its speed remains unchanged although gravity acts on it. (2 marks)
- (ii) A student says the Moon is much less massive than Earth and therefore exerts negligible force on Earth. Comment. (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
4. (a) (i) Rotation rate = $\frac{\omega}{2\pi} = \frac{5.0}{2\pi}$ = 0.795775 (rev s ⁻¹) $\approx$ 0.80 (rev s ⁻¹)	1M/1A	Accept: 0.79 ~ 0.80 (rev s ⁻¹)
(ii)  F_c correctly indicated.	1A	
$F_c = mr\omega^2$ = (0.03)(1 $\times$ cos 23.1°)(5.0) ² = 0.689866 N $\approx$ 0.690 N ($F_c = 0.7033402$ N $\approx$ 0.703 N for $g = 10$ m s ⁻²)	1M 1A	OR $T\cos\theta = F_c$ and $T\sin\theta = mg$ 1M $F_c = \frac{mg}{\sin\theta} \cos\theta = 0.689866$ N 1A Accept: $F_c = 0.70$ N
(iii) <u>Horizontal component</u> of tension provides the centripetal force, thus tension is <u>larger than</u> the centripetal force. OR $T\cos\theta = F_c \Rightarrow T > F_c$	1M 1A	$T\sin\theta = mg$ 1M $T = 0.750$ N $T > F_c$ 1A
(b) (i) The gravitational force is perpendicular to the moon's motion/displacement/velocity, thus no work is done on the moon by this force (k.e. unchanged)	1A 1A	
(ii) (The claim is incorrect) as, by <u>Newton's third law</u> of motion, gravitational force of the <u>same magnitude</u> (but in opposite direction) is acting on the Earth by the moon.	1A 1A	Accept: action and reaction pairs

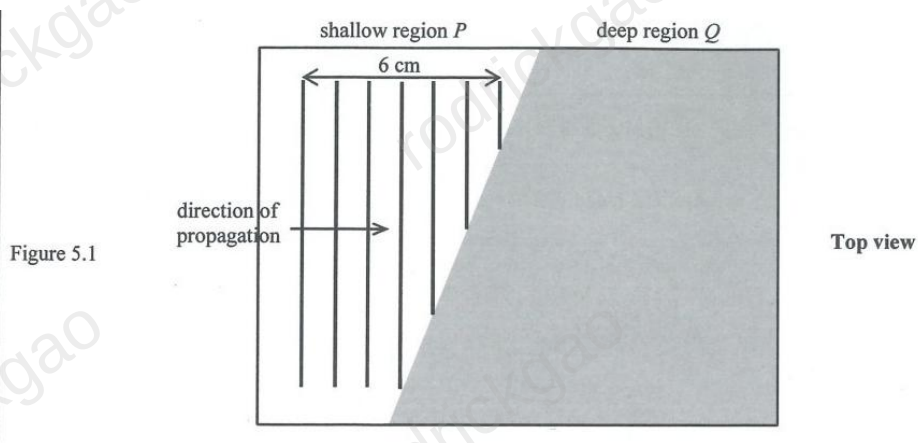
(a)(i) Rotation rate = $\omega/(2\pi) = 5.0/(2\pi) = 0.80$ rev s⁻¹.(ii) With radius $r = 1 \cos 23.1^\circ$, $F_c = mr\omega^2 = (0.03)(\cos 23.1^\circ)(5.0^2) = 0.690$ N.(iii) Only the horizontal component of tension supplies F_c while the vertical component balances mg , so $T > F_c$.

(b)(i) Gravity is perpendicular to the Moon's instantaneous velocity, so it does no work and changes direction but not speed.

(ii) The claim is false: by Newton's third law the Moon and Earth exert equal and opposite gravitational forces.

Question 5

A ripple tank has shallow region P and deep region Q. Straight water waves of frequency 10 Hz travel in P. The separation between seven crests in P is 6 cm.



- (a) (i) Find the wavelength in P. (1 mark) (ii) Find the wave speed in P. (1 mark)
- (b) In Q the wavelength is double that in P. (i) State the frequency in Q. (1 mark) (ii) Sketch the wave pattern in Q. (2 marks) (iii) Name the phenomenon at the boundary and explain its cause. (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
5. (a) (i) wavelength $\lambda = \frac{0.06}{7-1}$ $= 0.01 \text{ m} (= 1 \text{ cm})$	1A 1	e.c.f. from (a)(i)
(ii) speed $v = f\lambda = 10 \times 0.01$ $= 0.1 \text{ m s}^{-1} (= 10 \text{ cm s}^{-1})$	1M/1A 1	
(b) (i) frequency = 10 Hz	1A 1	'remain unchanged' is NOT accepted
(ii)		
shallow region P deep region Q	1A	Straight wavefronts <u>bend away from the normal</u> of the shallow-deep boundary. All correct with <u>double (longer) wavelength</u> having wavefronts <u>continuous at boundary</u>
6 cm	1A	
	2	
(iii) Refraction. It is due to the change in wavelengths/wave speeds in different media/depth.	1A 1A 2	Correct spelling

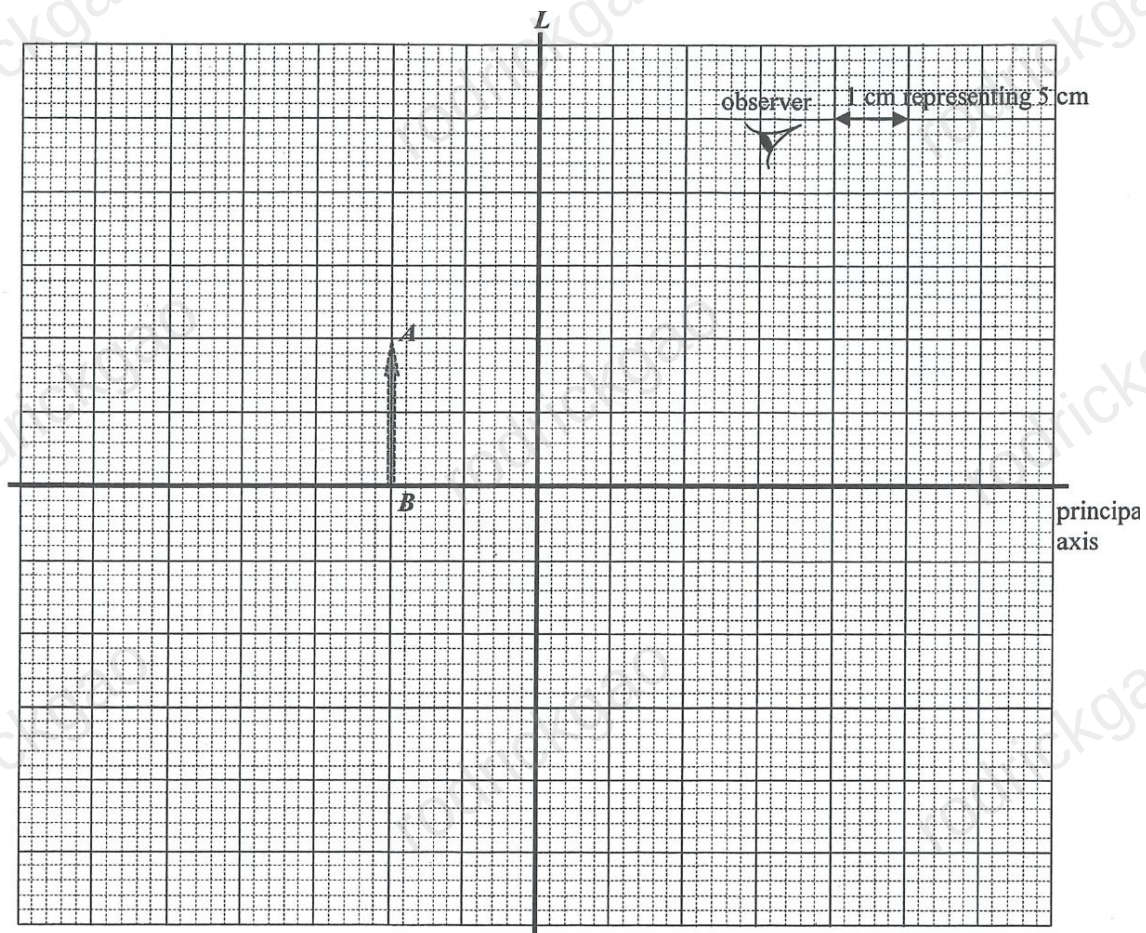
(a)(i) Seven crests span six wavelengths, so $\lambda = 6 \text{ cm}/6 = 1.0 \text{ cm} = 0.010 \text{ m}$.

(ii) $v = f\lambda = 10(0.010) = 0.10 \text{ m s}^{-1}$.

- (b)(i) Frequency remains 10 Hz across the boundary.
(ii) In the deep region draw continuous straight wavefronts with double spacing, bent away from the normal.
(iii) This is refraction, caused by the change of wave speed and wavelength with water depth.

Question 6

In Figure 6.1, AB is the virtual image of an object formed by lens L. The magnification is 0.4. The horizontal scale is 1 cm to 5 cm.



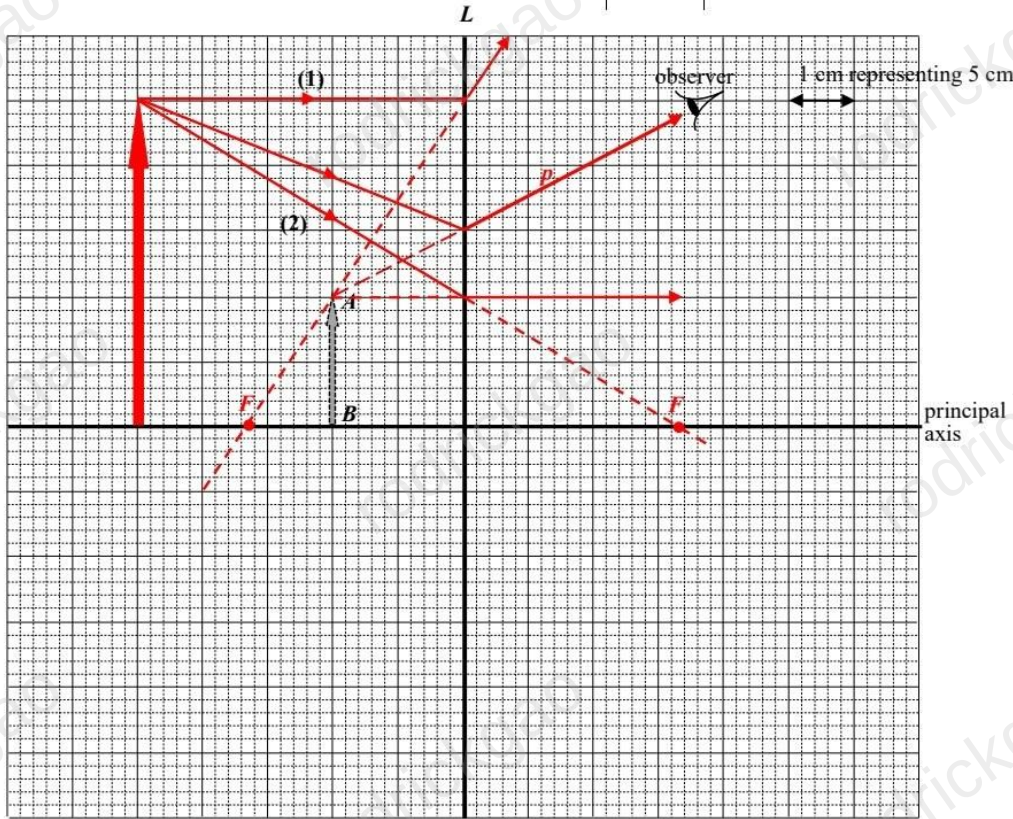
- (a) What kind of lens is used? Explain. (2 marks)
- (b) Indicate the position and height of the object. (2 marks)
- (c) By drawing a suitable ray, locate and mark focus F. Find the focal length. (3 marks)
- (d) Draw a ray from the object to show how the observer can see the tip A of the image. (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
6. (a) L is diverging/concave. Only diverging/concave lens forms <u>diminished, virtual image</u> .	1A 1A 2	This is the only accepted answer
(b)		
		
Correct position and height of object	2A 2	Correct position 1st mark All correct: 2 marks
(c) Correct ray to locate F and focus F correctly marked. Focal length = 16.5 cm (Accept 15.5 cm – 17.5 cm)	2M 1A 3	1st M mark: Ray (1) or (2) e.c.f. from (b) 2nd M mark: locate F
(d) Correct ray p from tip of object	2A 2	1st A mark: Diverging ray from the tip of the object, and can reach the observer 2nd A mark: The ray entering the eye comes virtually from the tip of the image

(a) L is a diverging (concave) lens because it forms an upright, diminished virtual image.(b) Since $m = \text{image/object} = 0.4$, place the object on the incident side with height 2.5 times the image and at the position fixed by the construction.(c) Extend a refracted ray backward to locate F ; the scale gives $f \approx 16.5$ cm.(d) Draw a ray from the object tip through the lens to the observer whose backward extension passes through image tip A .

Question 7

A battery has fixed e.m.f. ε and internal resistance r . Available apparatus includes resistors of known resistance R , a switch, an ideal voltmeter and connecting wires.

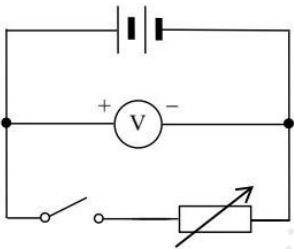
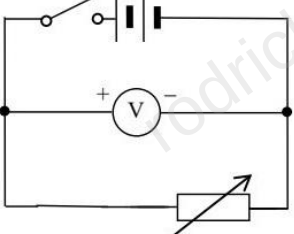
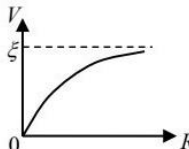
- (a) With a circuit diagram, describe an experiment to study how terminal voltage V depends on R . State ONE precaution. (5 marks)
- (b) Describe how V varies with R and express V in terms of ε , r and R . (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

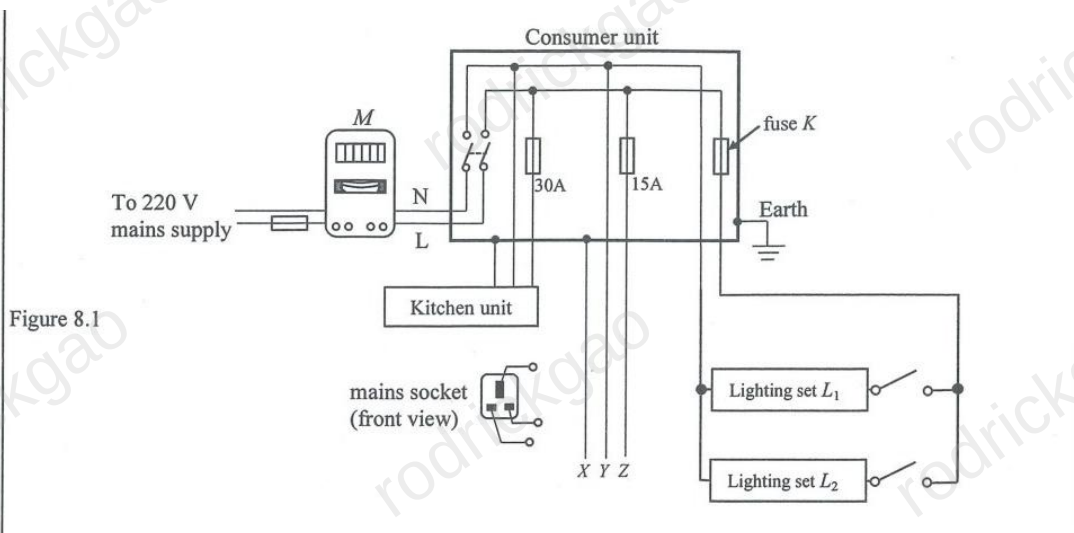
April 12, 2019

Solution	Marks	Remarks
(a)  Close the switch and record corresponding V and R readings Adjust the resistance R to lower/other value(s) and repeat the experiment Precaution: - First set the variable resistor to its maximum / a large value - Open the switch after each measurement - Any reasonable answer	1A 1A 1A 1A 1A 5	Correct circuit with correct symbol Correct polarity Alternative circuit  NOT accept Change of apparatus e.g. Using thicker connecting wires etc.
(b) Terminal voltage V delivered increases with increasing (loading) resistance R (or graphical representation) $V = \xi \frac{R}{R+r} \quad \text{OR} \quad V = \xi - \frac{\xi}{R+r} r$	1A 1A 2	Accept  NOT accept V is directly proportional to R V varies linearly with R

- (a) Connect R across the battery with a switch and measure terminal voltage V across the terminals. Start with maximum R , close the switch briefly, record V and R , then repeat for several lower R values; open the switch between readings.
- (b) V increases with R and approaches ε , with $V = \varepsilon R / (R + r)$.

Question 8

Figure 8.1 shows household wiring. The live L and neutral N mains wires connect through kilowatt-hour meter M to a consumer unit, where they branch into parallel circuits.



(a) Indicate how the mains socket should connect to X, Y and Z. (1 mark)

- (b) Lighting sets L_1 and L_2 , rated 300 W and 450 W, are connected in parallel to the branch with fuse K. (i) State one advantage of a parallel connection. (1 mark) (ii) From 3 A, 5 A, 10 A and 13 A, choose the most suitable fuse K and explain. (3 marks)
- (c) The kitchen unit includes:

Appliance	Rating	Effective operation per day
refrigerator	220 V, 500 W	8 h
electric kettle	220 V, 2000 W	0.5 h
induction cooker	220 V, 3000 W	2 h

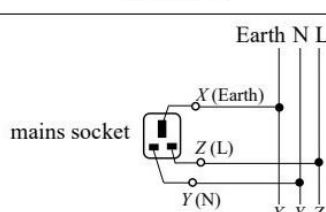
Find the daily cost if electrical energy costs \$0.9 per kW h. (3 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
8. (a) 	1A	
	1	
(b) (i) - If one of the lighting sets/circuits fails, the other (in parallel) can still operate, i.e. both work independently. - Both can work at the rated power. - Any reasonable answer	1A Any ONE	
	1	
(ii) $P = IV$ $(300 + 450) = I(220)$ $I = 3.409091 \text{ A} \approx 3.41 \text{ A}$ Thus 5 A fuse should be used.	1M 1A 1A	1M for substitution The unit of current can be omitted Correct deduction
	3	
(c) Electrical energy used per day $= 0.500 \text{ kW} \times 8 \text{ h} + 2 \text{ kW} \times 0.5 \text{ h} + 3 \text{ kW} \times 2 \text{ h}$ $= 11 \text{ kW h}$ Cost $= \$0.9 / \text{kW h} \times 11 \text{ kW h}$ $= \$9.9$	1M 1M 1A	
	3	

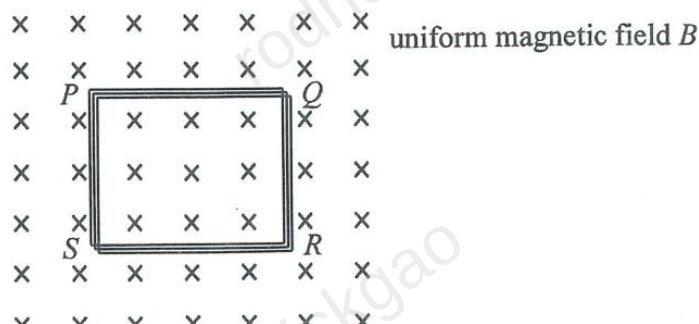
(a) Connect X to earth, Y to neutral and Z to live.

(b)(i) Parallel lighting sets operate independently and each receives the rated voltage.

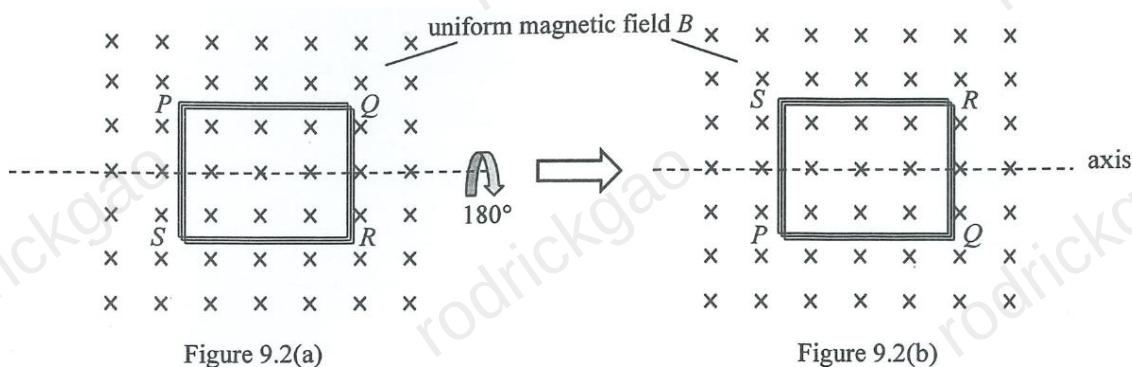
(ii) $I = (300+450)/220 = 3.41 \text{ A}$, so the smallest suitable rating above this current is 5 A.(c) Daily energy $= 0.5 \times 8 + 2 \times 0.5 + 3 \times 2 = 11 \text{ kW h}$; cost $= 11 \times \$0.9 = \9.9 .

Question 9

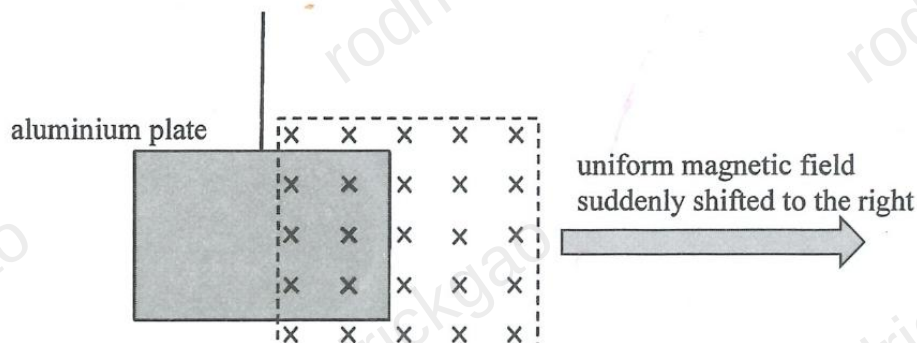
A rectangular coil PQRS has 20 turns, each of area 0.005 m^2 , in a uniform 0.3 T magnetic field directed into the paper.



- (a) The field strength decreases uniformly to zero in 0.5 s. (i) Explain why a current is induced. (2 marks) (ii) Calculate the change in total magnetic flux linkage and the induced e.m.f. (3 marks)
- (b) The coil is rotated uniformly through 180° in 0.5 s.



- (i) State the change in total magnetic flux linkage. (1 mark) (ii) At 90° , does current flow PQRS, PSRQ, or is there no induced current? (1 mark)
- (c) A thin rectangular aluminium plate is suspended partly inside a uniform field. The magnet is suddenly shifted to the right without touching the plate.



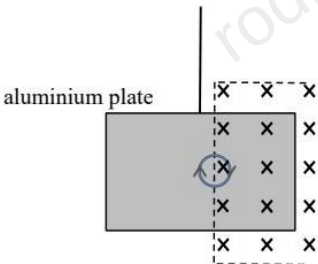
- (i) Mark where eddy currents are induced and show their direction. (2 marks) (ii) Describe the subsequent motion of the plate. (1 mark)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
9. (a) (i) By Lenz's law, <u>an e.m.f. would be induced</u> such that it opposes <u>the change</u> , i.e. <u>decrease of magnetic flux</u> (into the paper) by driving an induced current (clockwise) in the coil/circuit (complete).	1A 1A	1A for induced e.m.f. 1A for the change of magnetic flux
(ii) $N\Delta\Phi = NBA$ $= (20)(0.3)(0.005)$ $= 0.03 \text{ Wb}$ $\xi = \frac{N\Delta\Phi}{\Delta t} = \frac{0.03}{0.5}$ $= 0.06 \text{ V (or 60 mV)}$	1M/1A 1M 1A	Accept equivalent unit : $\text{T m}^2, \text{V s}$
(b) (i) The change of (magnetic) flux linkage is double that in (a)(ii), i.e. 0.06 Wb.	1M	e.c.f. from (a)(ii)
(ii) Direction of current : PQRS	1A	NOT accept clockwise/anticlockwise
(c) (i) 	1A 1A	Correct position (accept just within the magnetic field) Correct direction (clockwise) with complete circular path inside the aluminium plate
(ii) Move/swing to the right initially/momentarily/briefly.	1A	

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Examples:

	1A 1A		1A 1A
	0 1A		0 0
	1A 0		0 0
	0 1A		0 0
	0 0		1A 0

2019-DSE-PHY 1B-10

只限閱卷員參閱

FOR MARKERS' USE ONLY

Provided by dse.li

(a)(i) The decreasing flux induces an emf; by Lenz's law the clockwise current produces a field opposing the decrease.

(ii) $\Delta(N\Phi) = NBA = 20(0.3)(0.005) = 0.030 \text{ Wb}$ and $|\mathcal{E}| = 0.030/0.5 = 0.060 \text{ V}$.

(b)(i) Reversing the coil through 180° doubles the change to 0.060 Wb .

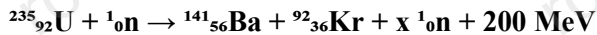
(ii) At 90° the induced current is in direction PQRS.

(c)(i) Draw a clockwise eddy-current loop within the part of the plate in the field.

(ii) The plate initially swings/moves to the right, following the magnet.

Question 10

(a) The nuclear-fission equation is:



(i) Find x . (1 mark) (ii) State a necessary condition for a fission chain reaction. (1 mark)

Evidence from Oklo, Africa indicates that natural fission occurred 2×10^9 years ago. Present ore contains 0.6% by mass of U-235.

(b) Given half-life of U-235 = 7.04×10^8 years, the sample data are:

Nuclide	2×10^9 years ago	Present
U-235	m_0 kg	0.060 kg (0.6%)
U-238	13.556 kg	9.940 kg (99.4%)

(i) Estimate m_0 . (2 marks) (ii) Determine whether natural fission was possible 2×10^9 years ago, given that at least 3% U-235 was required. (1 mark)

(c) Underground water slowed fast neutrons so that U-235 could capture them. The chain reaction stopped before U-235 concentration fell to 3%. Explain why. (2 marks)

Official marking scheme

只限閱卷員參閱

FOR MARKERS' USE ONLY

April 12, 2019

Solution	Marks	Remarks
10. (a) (i) $x = 3$	1A	Accept: exceed critical mass, suitable concentration of U-235
	1	
(ii) More neutrons are produced in each fission for triggering further fissions, i.e. $x > 1$.	1A	
	1	
(b) (i) $m = m_0 e^{-kt}$ $k = \frac{\ln 2}{t_{1/2}} = 9.846 \times 10^{-10} \text{ yr}^{-1}$ $0.06 = m_0 e^{-\ln 2 \times \left[\frac{2 \times 10^9}{7.04 \times 10^8} \right]}$ $m_0 = 0.429882832 \text{ (kg)} \approx 0.430 \text{ (kg)}$	1M 1A	OR $0.06 = m_0 \left(\frac{1}{2} \right)^{\frac{2 \times 10^9}{7.04 \times 10^8}}$
	2	
(ii) $\frac{0.430}{13.556 + 0.430} = 0.03073691 \approx 3.1 \% > 3\%$ Thus natural nuclear fission was possible.	1M/1A	
	1	e.c.f. from (b)(i)
(c) Underground water might run dry. OR Energy released by fission dries up the underground water.	1A	
Therefore, fission might stop without slow neutrons.	1A	
	2	

- (a)(i) Nucleon-number conservation gives $236 = 141 + 92 + x$, so $x = 3$.
- (ii) A chain reaction requires more than one neutron per fission to trigger further fissions, together with sufficient U-235/critical mass.
- (b)(i) $0.060 = m_0(1/2)^{2.0 \times 10^9 / 7.04 \times 10^8}$, giving $m_0 = 0.430$ kg.
- (ii) The earlier mass fraction was $0.430 / (13.556 + 0.430) = 3.07\% > 3\%$, so natural fission was possible.
- (c) Fission heating could boil away the underground water moderator; without slowed neutrons the chain reaction stopped.